Examiner only

7. *(a)* (i) A group of students were given three water samples. One was soft water, one was temporary hard water and one was permanent hard water.

Describe a method that the students could use to find out which is which. Give the expected results. You do **not** need to include the detail required to ensure a fair test. [4]

....

....

....

....

....

....

....

....

(ii) The samples from part (i) were labelled **X**, **Y** and **Z** and sent to have their ion content measured. The table shows the results.

Ions present	Ion content in water sample (mg/dm^3)		
	X	**Y**	**Z**
sodium	29	116	23
potassium	12	15	11
magnesium	31	4	98
calcium	141	2	27
hydrogencarbonate	30	19	219
chloride	17	14	20
sulfate	346	6	27
nitrate	12	15	19

Use this information to find whether sample **Z** is the soft water, the temporary hard water or the permanent hard water. Give a reason for your choice. [2]

....

....

....

....

Examiner only

(b) (i) Permanent hard water can be softened by ion exchange. Explain how ion exchange works. [2]

....

....

....

(ii) Explain why the resin in ion exchange stops working after continued use. [1]

....

....

(c) A litre bottle of mineral water contains 184 mg of dissolved calcium sulfate, $CaSO_4$. Calculate the number of moles of calcium sulfate present.

Give your answer to **three** significant figures.

1 mg = 0.001 g [2]

$A_r(Ca) = 40$ $A_r(S) = 32$ $A_r(O) = 16$

Moles of calcium sulfate = ... mol

11